Facilities, Equipment and Other Resources

The proposed research is in pure mathematics and requires no laboratory facilities or specialized equipment. Its needs are office space, access to the mathematical literature, and an active research community, all of which are available at the host institution.

Host institution

The fellowship will be held in the Department of Mathematics at the Massachusetts Institute of Technology (MIT), with Professor Davesh Maulik as sponsoring scientist. Professor Maulik will be in residence throughout the fellowship and will meet with the PI weekly. The PI will have an office in the department.

Library and computing

The MIT Libraries provide access to the research literature in mathematics, including MathSciNet. The project needs no computing resources beyond a personal laptop.

Scientific environment

MIT has a large and active community in algebraic geometry and neighboring fields, including moduli theory, enumerative geometry, and geometric representation theory. The PI will take part in the department's algebraic geometry seminar and preprint seminar, which Professor Maulik has invited the PI to help organize. The greater Boston area adds several geometers with Hodge-theoretic interests, notably Mihnea Popa at Harvard University, and many seminars in algebraic geometry and related fields.

Professional development and outreach

The department offers an intensive teacher-training program for new postdoctoral researchers, as well as workshops on grant writing and undergraduate advising, in which the PI will participate. The Broader Impacts activities described in the Project Description will draw on local programs, including the MIT Summer Research Program, the Harvard–MIT Mathematics Tournament (HMMT), and Boston-area math circles.